PSYC1071: Introduction to Psychology 1
Animal Learning and Cognition

Lecture 5:
Memory 2 and
Intro to Associative Learning

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Animal Memory

Examples of animal memory

- Capacity
- Time
- Number
- Categories

Properties of animal memory

Events from immediate past. Current behaviour may be influenced by recent events

- Short term memory
  - Habituation
  - Radial arm maze
- Long term memory
  - Consolidation theories
  - Retrieval theories

Short-Term Memory

Habituation A reduction in responsiveness to a stimulus as a result of its repeated presentation.

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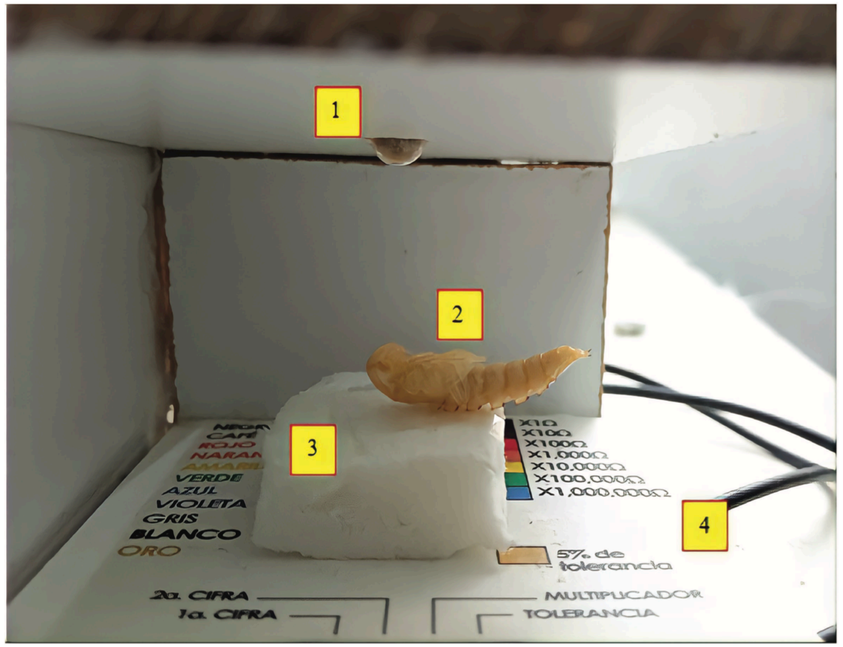

FIGURE 1 | Experimental chamber. 1 = LED lights, 2 = pupae, 3 = foam platform, and 4 = mini vibrating motor.

Bernal-Gamboa, Garcia-Salazar, & Gamez (2021)

Short-Term Memory

Habituation A reduction in responsiveness to a stimulus as a result of its repeated presentation.

Whitlow (1975) (rabbits) plethysmograph

S1		S2	
Tone 1	60 sec	Tone 1	
Startle!		weak startle	Habituation

Short-Term Memory

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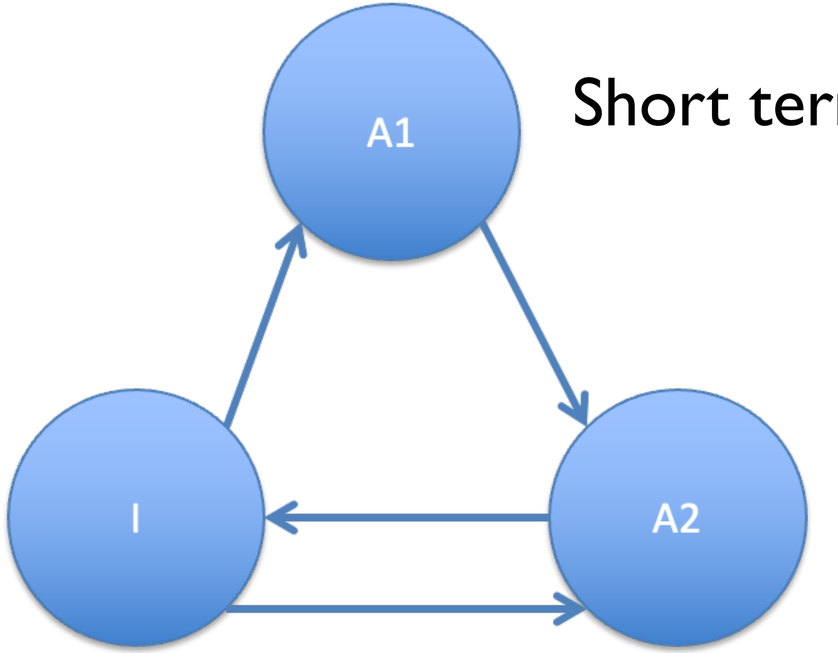
Startle!		weak startle
Tone 1	150 sec	Tone 1
Startle!		Startle!



Wagner (1976, 1981)

Short-Term Memory

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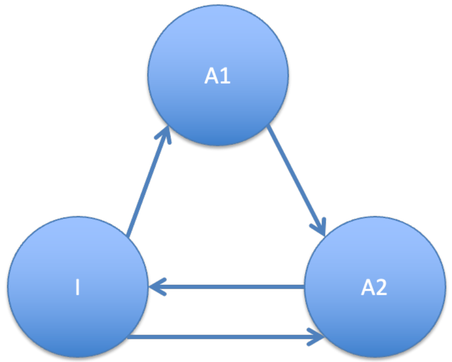


Effector Fatigue

Effector system - the physiological system responsible for producing a response

S1		S2
Tone 1	60 sec	Tone 1
Startle!		weak startle
Tone 1	60 sec	Tone 2
Startle!		Startle!

Receptor Fatigue

Receptor system - the physiological system responsible for perceiving the stimulus

S1		S2	
Tone 1	60 sec	Tone 1	
Startle!		weak startle	

Distractor

↓

Tone 1	60 sec	Tone 1	Dishabituation
Startle!		Startle!	

Retroactive Interference

Animal Memory

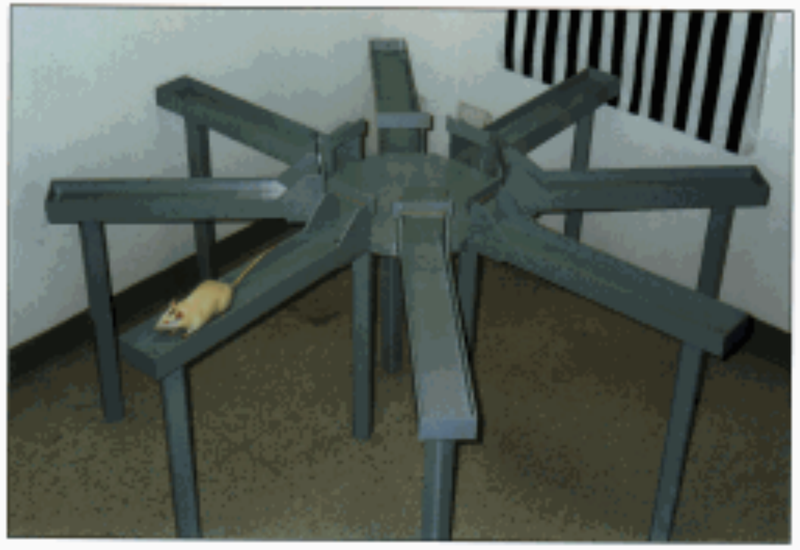
- Short-term memory
  - Habituation
    - Whitlow (1975)
    - Wagner (1976)
  - Radial arm maze
    - Beatty & Shavalia (1980)
    - Roberts (1981)
- Long-term memory
  - Consolidation theories
    - Electro-convulsive shock
  - Retrieval theories
    - Deweer, Sara & Hars (1980)

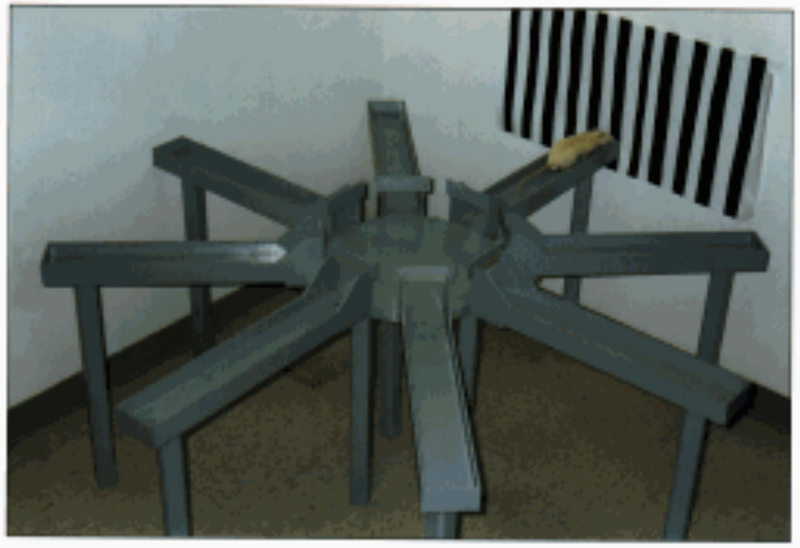
Short-lasting and susceptible to interference

Short-Term Memory

- Radial Maze Olton (1978)
  - Capacity - at least 8 items
  - Duration - up to 4 hours
  - Content - landmarks

- Retroactive interference
  - Beatty and Shavalia (1980)
  - Roberts (1981)





Animal Memory

- Short-term memory
  - Habituation
    - Whitlow (1975)
    - Wagner (1976)

    } Short-lasting and susceptible to interference
  - Radial arm maze
    - Beatty & Shavalia (1980)
    - Roberts (1981)

    } Lasts hours and less susceptible to interference
- Long-term memory
  - Consolidation theories
    - Electro-convulsive shock
  - Retrieval theories
    - Deweer, Sara & Hars (1980)

Short-Term Memory Summary

- STM occurs in variety of tasks
- STM is decaying memory trace (Wagner)
- Properties of STM vary according to task

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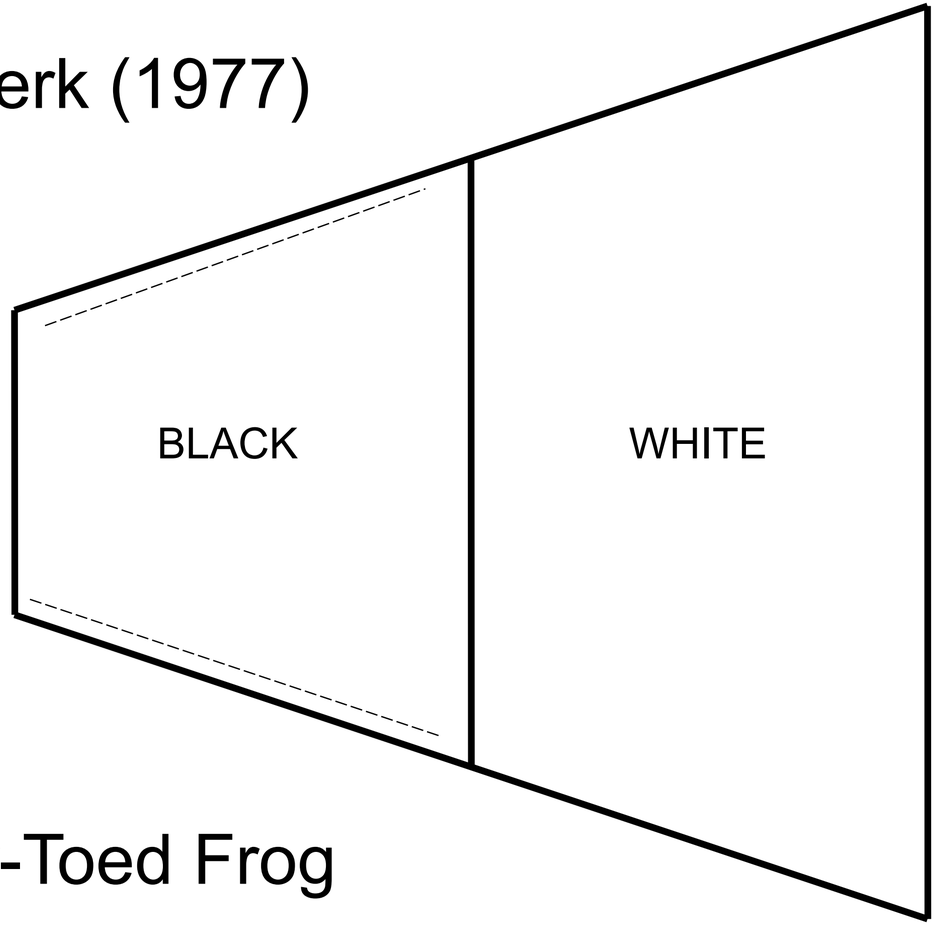
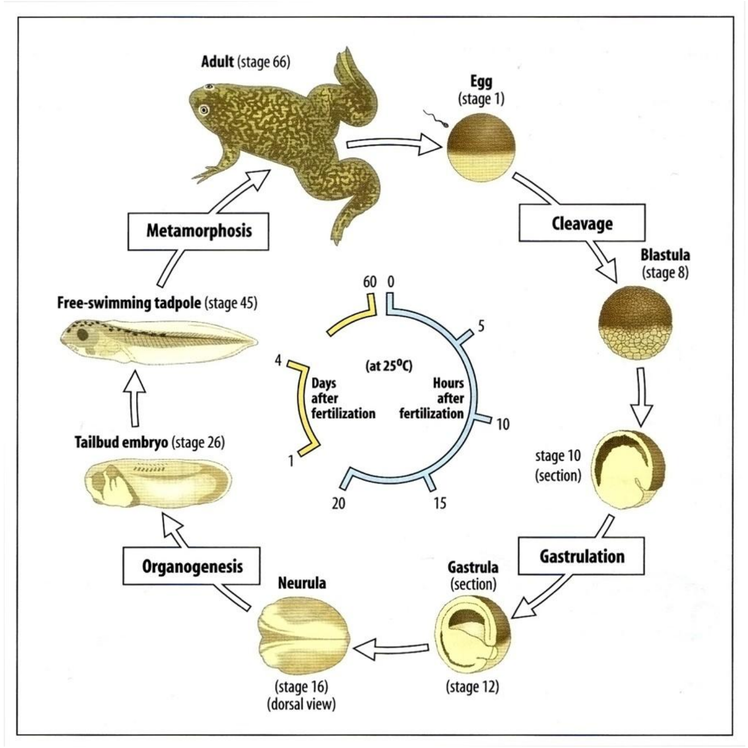
Miller and Berk (1977)



African Claw-Toed Frog

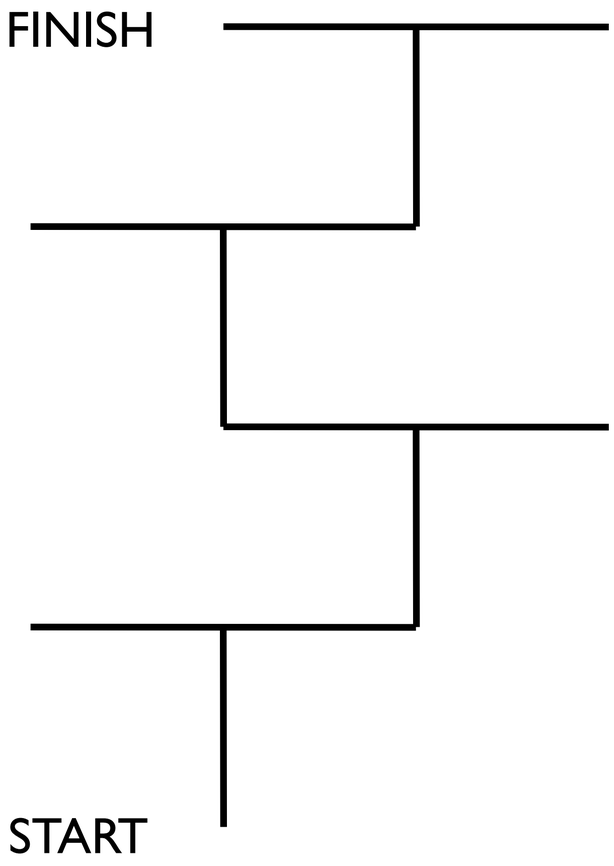


Mechanisms of LTM

- Consolidation
- Retention
- Retrieval

- **Consolidation Theories**
  - Hebb (1949)
  - Duncan (1949)
  - Electroconvulsive shock (ECS)

- **Retrieval Theories**
  - Deweer, Sara, & Hars (1980)



Deweer, Sara & Hars (1980)

Train = 30 sec 25 days Test = 150 sec

Train = 30 sec 25 days 90-sec Reminder Test = 40 sec

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Associative Learning

- What is learned in Pavlovian conditioning?
- What determines the nature of the conditioned response?
  - Stimulus substitution
    - Autoshaping
    - Food vs water
  - Preparatory response
    - Conditioned suppression

Associative Learning

Holloway and Domjan (1993)

CS	→	US	→	UR
Light		Receptive female		Approach

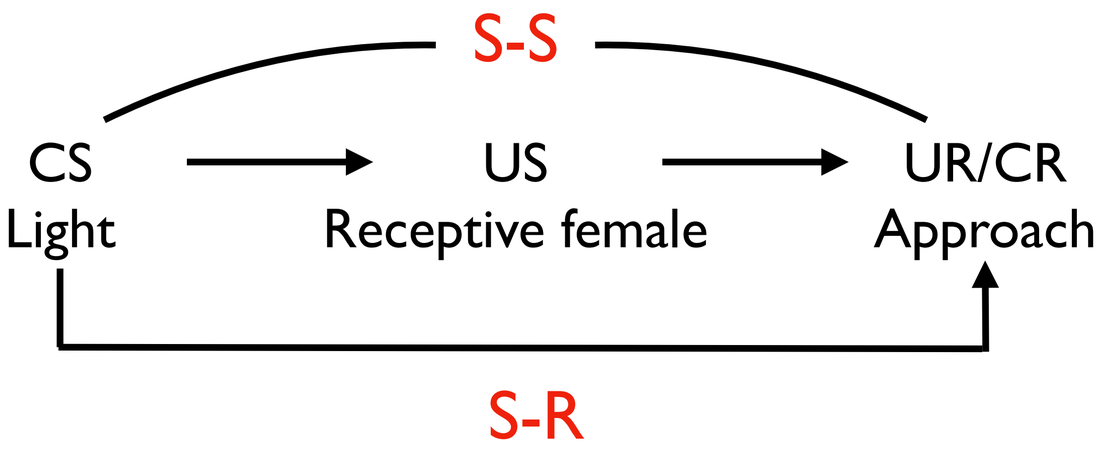
Associative Learning

Holloway and Domjan (1993)

CS	→	US	→	UR/CR
Light		Receptive female		Approach

Associative Learning

Holloway and Domjan (1993)

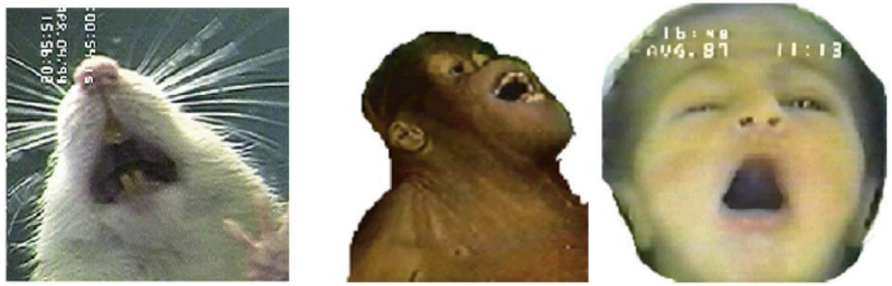


Holland (1990)

Stage 1	Stage 2	Test
Tone → Wintergreen+sucrose	Wintergreen + sucrose → LiCl	Tone + sucrose
Noise → Peppermint+sucrose		Noise + sucrose

Hedonic Reactions (sweet)

Aversive Reactions (bitter)



Holland (1990)

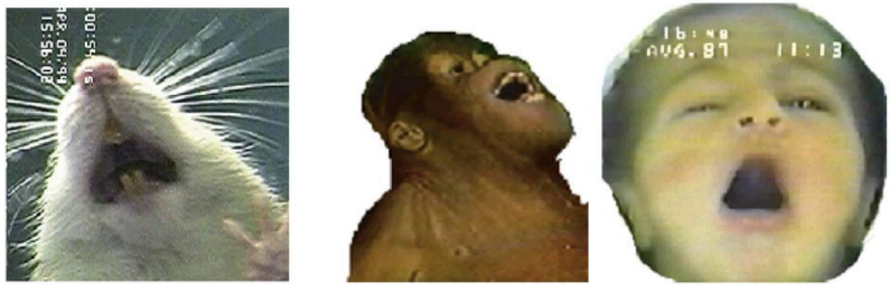
Stage 1	Stage 2	
Tone → Wintergreen+sucrose Noise → Peppermint+sucrose	Wintergreen + sucrose → LiCl	Tone + sucrose Noise + sucrose

Wintergreen
Illness

Peppermint

Hedonic Reactions (sweet)

Aversive Reactions (bitter)



The influence of the US on the CR

Holland

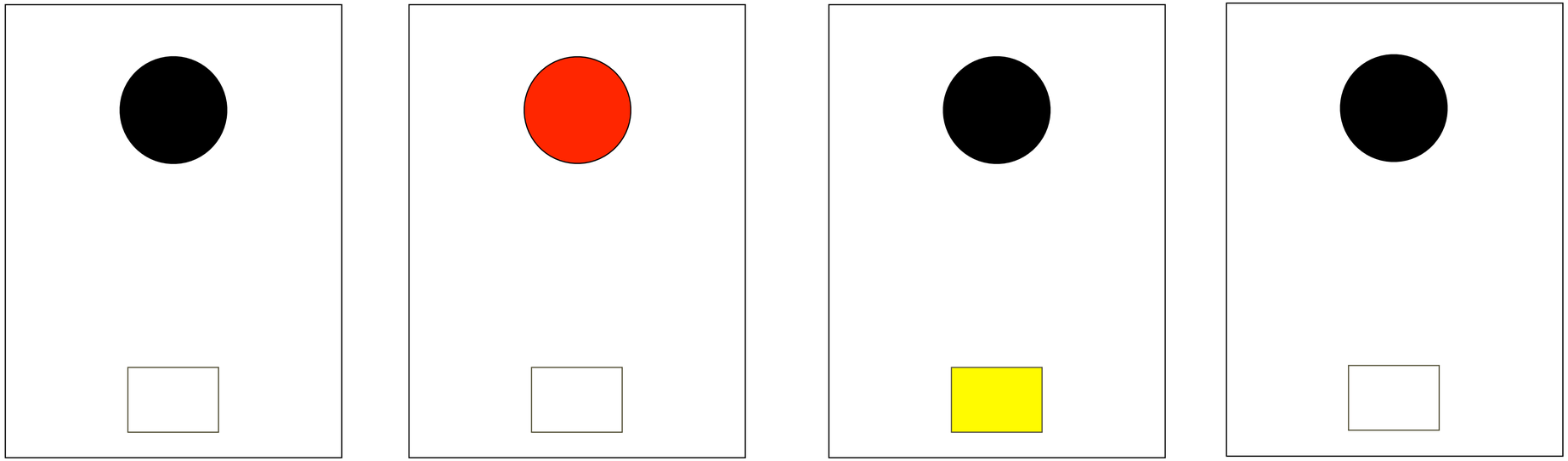
Konorski

USs have Specific and Affective qualities

Affective qualities can be Appetitive or Aversive

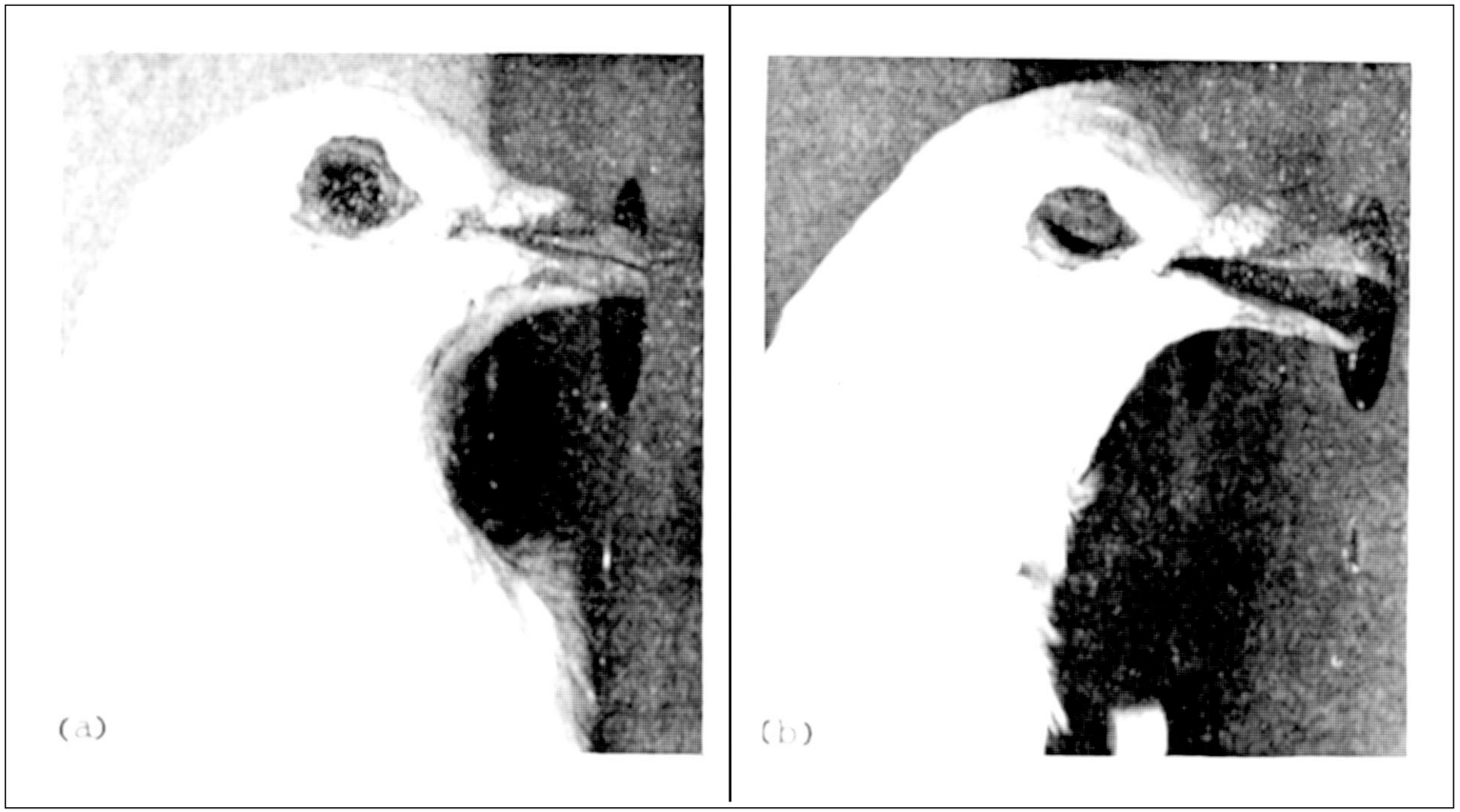
Stimulus Substitution

Autoshaping



ITI = 60 s CS = 5 s US = 5 s ITI = 60 s

ITI = intertrial interval



US is water

US is food

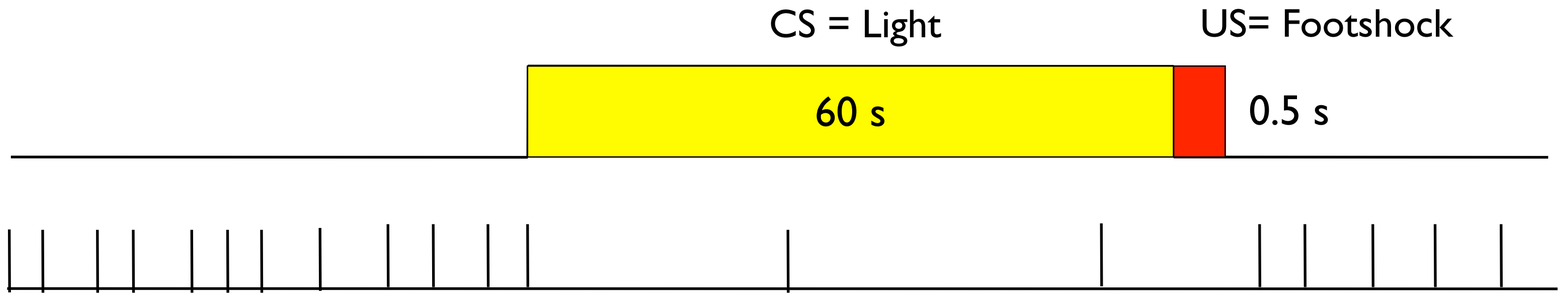
Preparatory Response

Salivation during CS that signals US

- general increase in activity

Conditioned Suppression

- decrease in movement



Lever Pressing

Associative Learning

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Next time:
Compensatory conditioned responses
The conditions of learning